

Priyank Jayraj

Information Retrieval & Applied Machine Learning

PhD Research Scholar, BITS Pilani



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EDUCATION

BITS Pilani – Ph.D., Computer Science & Information Systems, GPA 9.03/10

Aug 2023 – Present

BITS Pilani – B.E. Mechanical Engineering & M.Sc. Mathematics, GPA 6.53/10

Aug 2016 – May 2021

IBM Certified Data Science Professional

SKILLS

Languages Python, C/C++ (5+ yrs) · Java (Spring Boot), JavaScript (Node.js) (1+ yr)

ML / IR PyTorch, TensorFlow, CatBoost, LangGraph, Guardrails, LLMs, RAG, Graph Neural Networks

Systems AWS, GCP, Kafka, Airflow, CI/CD, Git

EXPERIENCE

Microsoft

Jun 2026 – Aug 2026

Bengaluru

Applied Scientist Intern

- Mapped the Power Pages maker journey across Studio, CLI, and VS Code. Isolated friction and drop-off for website creators, then turned those findings into action items for product and engineering.
- Built live dashboards so the intelligence team can see production numbers on performance and ease of use.
- Built a CatBoost model scoring Dynamics 365 customers for Power Pages adoption from revenue, industry, and usage. Recovered **88.5% of true natural adopters** on a held-out set for sales outreach.

ShareChat

Jan 2022 – Aug 2023

Bengaluru

Software Development Engineer I

- Built and ran backend systems—REST APIs, services, and job schedulers—for payments and gaming. Co-owned 6–8 production microservices.
- Cut infrastructure cost with a custom GCP Pub/Sub wrapper and compression on high-volume event streams.
- Used Pub/Sub, Cloud Functions, Cloud Scheduler, and the wider GCP stack to keep backend operations moving.
- Gave organization-wide engineering talks after finding high-leverage optimizations in web-service log streams.

PowerUp Money

May 2021 – Jan 2022

Bengaluru

Software Development Engineer I, formerly UniCards

- Built backend systems on AWS—REST APIs, services, and schedulers—for core product features.
- Owned key microservices on credit-card decisioning and risk/fraud detection.
- Joined in a very early phase of the startup and often shipped several features in a day.

RESEARCH

Ministry of Tribal Affairs · BITS Pilani

Aug 2024 - Present
Pilani

PhD Researcher, Department of CSIS

- Building pipelines that turn welfare-scheme documents into posters and animations, so low-resource language communities can understand the schemes in a culturally faithful way.
- Designed Coherent Thematic Unit (CTU) segmentation, then **CTU-FlowRAG**: role-conditioned retrieval over typed sentence graphs. On 3,048 scheme documents (187,220 sentence units): 0.82 nDCG@10, 0.75 MRR@10, 0.96 role coverage.
- Working on iterative generation of connected images so character and style stay consistent across a poster sequence.
- Built **QCCAF**, a query-conditioned RAG fusion layer that adds local cultural context before image prompts are written. 0.93 nDCG@10 on a new Santali IR benchmark; zero-shot transfer to Khasi, Bodo, and Garo.
- Proposed **Bandit-Quota**, a latency-budgeted contextual bandit for per-query retriever selection. 2.4-9.3× faster than a full ensemble on seven BEIR datasets, matching or beating quality on six of seven.

PUBLICATIONS

All with Poonam Goyal and Navneet Goyal.

- [1] **Bandit-Quota: Multi-Retriever Selection under Latency Budgets via Contextual Thompson Sampling.**
ACM International Conference on Information and Knowledge Management (CIKM), 2026. **Accepted.**
- [2] **Role-Conditioned Structured Retrieval on Typed Sentence Graphs for Welfare-Policy Documents.**
IEEE International Conference on Data Science and Advanced Analytics (DSAA), 2026. **Accepted.**
- [3] **QCCAF: A Cultural Retrieval Benchmark and Query-Adaptive Fusion for Tribal Welfare Communication.**
IEEE International Conference on Data Science and Advanced Analytics (DSAA), 2026. **Accepted.**

TEACHING

CSIS, BITS Pilani
Teaching Assistant
Aug 2023 - Present

Helped design courses and ran lectures and labs for NLP, Generative AI, and Advanced Algorithms.

COMPETITIVE PROGRAMMING

Codeforces **1813**
CodeChef **2051**
1000+ problems solved

ACM ICPC
2021 India online – 52
2021 Asia Pune – 142
2020 Asia Kanpur – 52
2020 India online – 214
2019 Asia Kharagpur – 54
2019 Asia Kanpur – 36
2019 India online – 80

SOCIETIES

BITSACM
Core team

CPSIG
Coordinator